Swarm Scaling and Network Dynamics with Small Reasoning Models

Experimental plan: single agents, silent ensembles, and freely communicating swarms on Qwen3 open-weight models

Draft plan

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Abstract

We plan to measure how the accuracy of small open-weight reasoning models on competition math scales with token budget, in three settings: one agent thinking longer, N independent agents combined by plurality vote, and N agents that communicate freely while drawing on one shared token pool. We record an exact per-agent breakdown of where tokens go, and we track the communication network and answer dynamics that emerge. From these curves we estimate the parallelizability (Ord) exponent λ, defined by the statement that N agents achieve as much as one agent working N^{λ} times as long.

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1 Goals and questions

In this computational experiment, we attempt to answer the following questions.

Q1. Single-agent scaling. How does accuracy grow with the number of reasoning tokens one agent spends?

Q2. Communicating swarms. Do N agents that communicate with each other beat a single agent at a given token budget?

Q3. Emergent dynamics. Are there network or social dynamics that emergently arise from communicating swarms?

2 Models and serving

2.1 Tested models

We use the Qwen3 family in thinking mode. All original sizes share one tokenizer, one chat template, and the same `<think>...</think>` convention, so the accounting code is identical across sizes.

- **Primary:** Qwen3-4B (thinking mode). Cheap enough for many agents, and far from saturation at low budgets.
- **Capability axis:** Qwen3-1.7B and Qwen3-8B through the identical harness. This tests whether λ or the swarm gain depends on model strength.

2.2 Sampling and context windows

- Generation: temperature 0.6, top-p 0.95, top-k 20, min-p 0, as recommended for thinking mode. No greedy decoding during generation, since it causes repetition. Each agent and each trajectory gets its own seed. Sampling noise is the only source of diversity among identical agents, so these settings are fixed across all conditions.

- Context: 32,768 tokens natively; up to about 131k with YaRN. Open-source servers implement static YaRN, which can slightly hurt short inputs. We use the *same* context setting in every condition.

2.3 Serving

We use vLLM: prefix caching on, the Qwen3 reasoning parser (splits reasoning from content), and the Hermes-style tool-call parser.

Chat template. By default the Qwen3 template removes earlier thinking from the conversation history. An agent therefore keeps only what it wrote visibly. Template versions differ in how they treat thinking inside tool-call loops. We render one full agent context with `apply_chat_template`, read it, and pin the template file.

KV-cache budget. The cache, not the weights, limits N. Per token, the cache size is

$$m_{\text{KV}} = 2 \times n_{\text{layers}} \times n_{\text{KV heads}} \times d_{\text{head}} \times \text{bytes per value}. \tag{1}$$

For 36 layers, 8 KV heads, $d_{\text{head}} = 128$ and bf16 this gives $\approx 144\,\text{KiB}$ per token, so a full 32k context needs $\approx 4.5\,\text{GiB}$ per agent, and 16 agents need $\approx 72\,\text{GiB}$. We read the exact values from each model's config file. Prefix caching only shares the common system prompt and problem text, not each agent's own reasoning.

3 Benchmarks

- **AIME:** answers are integers in $\{0, \ldots, 999\}$; grading is exact match on the boxed answer. Chance level is essentially zero.
- **AMC 10/12:** five answer options, so chance is 20%, and wrong answers collide by accident, which can look like consensus. We either remove the options and grade as free response, or keep them and report chance-corrected accuracy $(A - 0.2)/(1 - 0.2)$.
- **Contamination:** older AIME and AMC sets are likely in training data. We prefer problem sets published after the model's release date (for example AIME 2026), and we check dates explicitly.
- **Operating range:** a pilot measures per-problem solve rates $\hat{p}_i(b)$ (Section 5). We want most problems in the informative middle range, not near 0 or 1. If we select problems using pilot results, we use *fresh* samples for the main measurement, to avoid regression to the mean in the baseline.

Notation: problems $i = 1, \ldots, P$ with correct answers y_i.

4 Token accounting

4.1 Categories

Every token belongs to exactly one category.

Symbol	Category	Type
C_{think}	private reasoning inside the think block	generated
C_w	writing a post to the shared board	generated
C_{dms}	sending a direct message	generated
C_{ans}	setting the current answer	generated
C_h	reading board headlines	read (prefill)
C_x	expanding (reading in full) a board post	read (prefill)
C_{dmr}	reading a received direct message	read (prefill)
C_o	overhead: system prompt, tool schemas, template tokens, retries	mostly read

For agent i, the cost of one call is

$$C^{(i)} = \underbrace{C^{(i)}_{\text{think}} + C^{(i)}_{w} + C^{(i)}_{\text{dms}} + C^{(i)}_{\text{ans}}}_{\text{generated}} + w \underbrace{\left(C^{(i)}_{h} + C^{(i)}_{x} + C^{(i)}_{\text{dmr}} + C^{(i)}_{o}\right)}_{\text{read}}. \tag{2}$$

The weight w sets the currency. In FLOPs, generating and reading a token cost about the same ($\approx 2\times$ active parameters), so $w \approx 1$. In wall-clock time or API price, reading is cheaper, so $w < 1$. We report both, plus elapsed time.

4.2 Exact measurement

- **Generated tokens.** `<think>` and `</think>` are single special tokens. We request token IDs from the server and split at the index k of `</think>`: tokens before k are C_{think}, tokens after it are visible output. Visible output is split into write, DM and answer spans by the tool-call structure. A trajectory with no closing tag was truncated.
- **Read tokens.** Every read token comes from a message our harness appends (headlines, expanded post, DM). We tokenize and label each message before sending it. With prefix caching, we count only newly ingested tokens, and we check this against the server's cached-token report.
- **Conservation check.** For each call, the category sum must equal the server-reported token count up to a few template tokens. A drifting residual signals a parsing error.
- **Storage.** We store token IDs, not only decoded text, because decoding and re-encoding do not always round-trip.

5 Experiment 1: single agent, accuracy versus budget

We assess the performance of a single agent systematically. To do this, we take into account the token consumption used to generate an answer to a single problem, which may or may not be correct. Let j be a replicate (seed) number. The answer of agent $\mathcal{A}$ to problem i in the benchmark set, at an imposed token budget b, is done through a probing scheme of the trajectory τ_{ij}. Each trajectory τ_{ij}, $j = 1, \ldots, K$ (typically $K = 32$–64) for problem i has a token cap

$$B_{\max} = W - |\text{prompt}| - R, \tag{3}$$

where W is the context window and R is a reserve of a few hundred tokens for the probe suffix and answer. For each trajectory we record the natural prefill-decode token spend L_{ij}.

The answer of agent $\mathcal{A}$ by probing trajectory τ_{ij} at budget b is then

$$\mathrm{Ans}_{\mathcal{A}}(\tau_{ij}, b) = \begin{cases} \text{final answer} & \text{if } L_{ij} < b \text{ (thinking already finished),} \\ \mathrm{decode}\big(\text{prompt} \oplus \tau[:b] \oplus s_{\text{stop}}\big) & \text{otherwise.} \end{cases} \tag{4}$$

Here, $\oplus$ denotes concatenation, s_{stop} is a short stop-thinking message prompt, followed by `</think>` and the start of a boxed answer, in the style of Qwen3's thinking-budget mechanism. This probing mechanism forces the agent to arrive at an answer if the token spend is greater than the budget. We use greedy decoding during this forcing case. Probe tokens are not counted in the token spend calculation.

5.1 Probe checkpoints

We probe at logarithmic spacing of token budget

$$b_m = b_0\, 2^{m/3}, \qquad m = 0, 1, \ldots, \qquad b_m \le B_{\max}, \quad b_0 \approx 256. \tag{5}$$

```
for t in trajs:                                    # token IDs from generation
    for b in checkpoints:
        if t.think_end is not None and t.think_end < b:
            ans = t.final_answer                   # answer already fixed
        else:
            fork = prompt_ids + t.ids[:b] + STOP_THINKING_IDS
            ans = decode(fork, max_tokens=12, temperature=0)   # discarded fork
        correct[t.problem, t.idx, b] = (parse_boxed(ans) == gold[t.problem])
```

5.2 Aggregation

Let $c_{ij}(b) = \mathbf{1}[\mathrm{Ans}(\tau_{ij}, b) = y_i]$ i.e., 1 if the answer of agent trajectory τ_{ij} within token budget b (where $L_{ij} < b$) is correct, 0 otherwise. We average first within each problem i with K replicates (seeds) each, then over problems $i = 1, 2, \ldots, P$:

$$\hat{p}_i(b) = \frac{1}{K} \sum_{j=1}^{K} c_{ij}(b), \qquad\qquad A_1(b) = \frac{1}{P} \sum_{i=1}^{P} \hat{p}_i(b). \tag{6}$$

This weights every problem equally, regardless of how many tokens it consumed.

The x-axis is realized spend, not nominal budget. Trajectories that finish early stop spending. The tokens spent at budget b, considering that the agent can solve the problem below the imposed budget, are

$$s_1(b) = \frac{1}{PK} \sum_{i,j} \min\big(L_{ij},\, b\big) \;\le\; b. \tag{7}$$

We plot A_1 against s_1 on a logarithmic axis. This is the quantity comparable to a swarm's pooled spend.

5.3 Toy example

Replicate	L	final	$b = 5$k	10k	20k	30k
r_1	20k	wrong	0	0	0	0
r_2	18k	correct	0	0	1	1
r_3	9k	correct	0	1	1	1
r_4	25k	wrong	0	0	0	0
r_5	12k	correct	0	0	1	1
$\hat{p}_A(b)$			0	0.2	0.6	0.6

Realized spend for A at $b = 30$k is $(20 + 18 + 9 + 25 + 12)/5 = 16.8$k, not 30k; at $b = 20$k it is $(20 + 18 + 9 + 20 + 12)/5 = 15.8$k. Across problems:

Problem	5k	10k	20k	30k
A	0	0.2	0.6	0.6
B (easy)	0.4	0.8	1.0	1.0
C (hard)	0	0	0.2	0.2
D (never solved)	0	0	0	0
$A_1(b)$	0.10	0.25	0.45	0.45

5.4 Reading the curve

- **Floor:** at very small budgets the forced answer is close to a guess.
- **Log-linear region:** fit $A_1 \approx \alpha + \beta \log_2 s_1$. The slope β is the gain per doubling of tokens.
- **Plateau:** may come from the model or from the cap $B_{\max}$. A cap-induced plateau favours swarms artificially, because a single agent cannot spend beyond its context window without compaction.
- Values at neighboring checkpoints share trajectories and are correlated. An answer can also flip back, so the curve need not be monotone.
- **Survivor effect:** if correct trajectories tend to finish early, then trajectories still running at large b are enriched for hard or looping cases. This bends the curve and is another reason to use realized spend.

6 Experiment 2: silent ensembles

This experiment needs no new generation. It reuses the K trajectories of Experiment 1.

Plurality vote. For problem i, draw a subset of replicates S of size N without replacement. The ensemble answer at budget b is the most frequent answer among $\{\mathrm{Ans}(\tau_{ij}, b) : j \in S\}$, with ties broken uniformly at random. We estimate

$$\hat{p}^{\mathrm{vote}}_{i,N}(b) = \frac{1}{Q} \sum_{q=1}^{Q} \mathbf{1}\big[\,\mathrm{plurality}_{S_q}(b) = y_i\big], \qquad A^{\mathrm{vote}}_N(b) = \frac{1}{P} \sum_i \hat{p}^{\mathrm{vote}}_{i,N}(b), \tag{8}$$

with Q random subsets per problem. Here resampling replicates is correct, because the within-problem sampling is the quantity being estimated. The spend is $s_N(b) = N\, s_1(b)$ in total, and $s_1(b)$ per agent (which sets wall-clock time).

7 Experiment 3: freely communicating swarm

7.1 Environment

N agents work on one problem at a time. Each agent uses the following tools:

Tool	Effect
`post(text)`	append a message to the shared board
`read_headlines()`	list new posts: author ID and first line
`expand(post_id)`	read one post in full
`dm(agent_id, text)`	send a private message to another agent
`read_inbox()`	read received private messages
`set_answer(a)`	update the agent's current declared answer

The two-stage read (headlines, then expand) makes attention observable: each `expand` is a directed, weighted edge $i \to j$.

Rules.

- **Shared pool.** All agents draw on one pool of B_{pool} tokens. A per-call generation cap $g_{\max}$ stops one agent from spending the pool in a single call.
- **Asynchronous execution.** Agents run concurrently. The clock is cumulative pool spend S.
- **Identity.** Agent IDs are short random strings, redrawn for every problem and seed. This prevents label bias such as deference to "Agent 1," and makes ID mentions easy to find in text.
- **Prompt.** One short, identical system prompt for all agents. It names the tools, the problem, N, and the stopping rule, and says nothing about cooperation, roles, or verification.
- **Termination.** The run stops when the pool is empty or all agents declare they are done. The swarm answer is the plurality of declared answers.
- **Grammar.** Constrained decoding enforces valid tool-call structure. The same grammar is used in Experiment 1 for fairness.

7.2 Causality bookkeeping

Each call has a start spend S_{start}, an end spend S_{end}, and the board version it saw at the start. Posts that arrive between S_{start} and S_{end} were invisible to that call. An influence edge is valid only if the read happened before the answer change. With real concurrency, an agent with short calls acts more often; we log call lengths to separate "talks a lot because it is influential" from "talks a lot because its calls are short." Runs are not exactly reproducible; a discrete-event scheduler on a virtual token clock is a fallback if reproducibility is needed.

7.3 Cost tensor and conservation

At probe checkpoints S_m we store the tensor

$$\mathcal{C}[m, i, c] = \text{cumulative tokens of agent } i \text{ in category } c \text{ up to } S_m, \tag{9}$$

and its increments between checkpoints.

- Every DM is paid twice: once by the sender (C_{dms}), once by the receiver (C_{dmr}).
- A post is written once but can be read up to $N-1$ times, so over any interval

$$\sum_i C_x^{(i)} \;\le\; (N-1) \sum_i C_w^{(i)}. \tag{10}$$

This is the origin of the N^2 communication cost. We track the *read amplification*

$$\rho(S) = \frac{\sum_i \left(C_x^{(i)} + C_{\text{dmr}}^{(i)}\right)}{\sum_i \left(C_w^{(i)} + C_{\text{dms}}^{(i)}\right)} \tag{11}$$

and the *communication fraction* $1 - \sum_i C_{\text{think}}^{(i)} / \sum_i C^{(i)}$. Increments reveal phases, for example thinking-heavy exploration followed by reading-heavy checking.

7.4 Probes in the swarm

At each checkpoint S_m and for each agent i we record:

- the **declared answer** $d_i(S_m)$ from `set_answer`;
- the **forced answer** $f_i(S_m)$: fork the agent's current context, including any unfinished reasoning, append the stop suffix, decode, and discard the fork.
- for five-option problems, the forced **belief vector** $\pi_i(S_m)$ from log-probabilities.

The forced-answer plurality is directly comparable to Experiments 1 and 2. The declared plurality is what the swarm says.

Conformity signal. The divergence fraction

$$D(S) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}\big[d_i(S) \neq f_i(S)\big] \tag{12}$$

counts agents whose public answer differs from their private one. Agents that declare a hub's answer while privately still holding another are conforming without being convinced. $D(S)$ is a candidate early warning for wrong consensus.

8 Estimating λ

Compare curves horizontally. Suppose N agents with per-agent spend b reach accuracy $A_N(b)$. Find the single-agent spend b' with $A_1(b') = A_N(b)$, interpolating A_1 linearly in $\log s$. Then, from $N^\lambda b = b'$,

$$\lambda = \frac{\log\left(b'/b\right)}{\log N}. \tag{13}$$

In the log-linear region with single-agent slope β_1 and pure swarm-scaling slope β_N (accuracy versus log of total tokens at fixed per-agent budget), this reduces to $\lambda \approx \beta_N/\beta_1$. We compute λ inside each bootstrap draw. If $b' > B_{\max}$, the single agent cannot reach that accuracy without compaction; we report such points separately rather than extrapolating.

We estimate λ separately for silent voting and for communicating swarms, for each model size, and for homogeneous and heterogeneous swarms.

9 Network and social dynamics

9.1 Graphs

Between consecutive checkpoints we build directed weighted graphs:

- **Attention graph:** weight W^{att}_{ij} = tokens agent i spent expanding posts by j.
- **Message graph:** weight W^{dm}_{ij} = DM tokens from i to j.
- **Mention graph:** occurrences of j's random ID in i's posts and in i's private reasoning. Mentions inside private reasoning show whom an agent is thinking about.

9.2 Observables

Phenomenon	Measure
Talkers, free-riders	Gini of per-agent spend, $G = \dfrac{\sum_{i,j} \lvert x_i - x_j \rvert}{2N^2 \bar{x}}$; share of the top agent
Hubs, leadership	in-strength centralization of W^{att}; persistence of the hub across checkpoints
Influence	adoption events: j switches to answer a shortly after expanding a post by i that asserts a; adoptions credited per source
Dialogue	reciprocity of DM edges; pairs versus star structure
Copying	echo fraction: n-gram overlap of each post with earlier posts
Consensus	answer entropy $H(S) = -\sum_a q_a(S) \log q_a(S)$, where q_a is the fraction of agents declaring a; fraction correct; time to 75% plurality
Herding failure	runs ending in confident consensus on a wrong answer; the network and adoption history before them

9.3 Controls and causal tests

- **Null models:** degree-preserving rewiring of observed edges; permutation of agent labels within time windows. A statistic counts as structure only if it beats these baselines.
- **Label check:** an agent's role must not be predictable from its ID or start order.
- **Replay interventions:** all contexts are saved, so we re-run sampled decisions with a post deleted or its claimed answer changed. If the adoption disappears, the post caused it. We do this on a sample of adoption events. Reasoning traces generate hypotheses; replays test them.
- **Attention attribution (optional, offline):** replay sampled contexts in Hugging Face Transformers with attention outputs and sum attention mass on each author's text spans.

- **Ensembles:** many seeds and problems, so we report distributions of emergent structure (how often a talker appears, and whether it helps or hurts accuracy) rather than single anecdotes.

10 Logging

Two tables are enough; everything else is derived offline.

Event log (JSONL, one row per event).

```
{"run": r, "problem": p, "agent": "k7q", "event": "expand",
 "target_agent": "m2x", "post_id": 118, "S_start": 60120, "S_end": 60532,
 "board_version": 41, "tok_gen": 0, "tok_read": 412,
 "answer_before": 204, "answer_after": 204, "pool_left": 61833}
```

Probe table (one row per checkpoint and agent).

```
{"S": S_m, "agent": "k7q",
 "cum": {"think": 9120, "write": 640, "dms": 210, "dmr": 180,
         "headlines": 950, "expand": 3310, "ans": 12, "overhead": 1400},
 "declared": 204, "forced": 117, "belief": null, "in_flight": true}
```

We also store raw token IDs for every call, and log-probabilities where affordable.

11 Pilot and compute

1. **Pilot** ($K = 16$–32, Qwen3-4B): measure natural lengths L, set $B_{\max}$ near the 95th–99th percentile of L, check that the forced-answer suffix works, and measure $\hat{p}_i(b)$ to choose the problem set and budget range.
2. **Compute estimate** for Experiment 1: total generated tokens $\approx P\,K\,\bar{L}$. For example, $30 \times 64 \times 15{,}000 \approx 2.9 \times 10^7$ tokens. The pilot replaces $\bar{L}$ with the measured value. Probe cost is small with prefix caching but should be measured.
3. **Sanity check** for Experiment 3: a swarm of $N = 1$ must reproduce Experiment 1 apart from tool overhead. The difference is the fixed cost of the communication environment, and it explains why swarms lose at small budgets.

12 Open design decisions

1. **Visibility of the pool.** Can agents see the remaining pool and each other's spending? Seeing the pool may cause a rush to spend; seeing peers' spending lets agents react to talkers; hiding both is the least imposed choice.
2. **Memory of own reasoning.** Keep the default template (past thinking removed; the board becomes the only memory) or retain each agent's own thinking (higher read cost, possibly more looping).
3. **Main figure axis.** Fixed model with a budget sweep, or fixed budget with a model-size sweep.
4. **Benchmark.** AIME 2026 versus AMC as free response, chosen from the pilot so that $\hat{p}_i$ lies mostly in the middle range.
5. **Context.** 32k native with compaction, or YaRN in all conditions.

Key risks

- Small models may make malformed tool calls or loop; constrained decoding and the loop signature in traces mitigate this.
- Small models may agree too readily; this is both a risk to accuracy and a phenomenon to measure through $D(S)$ and adoption events.
- Contamination can inflate single-agent accuracy and compress the gains we want to measure.
- Hardware load changes the order of events in asynchronous runs; logging call lengths and, if needed, a virtual clock address this.